

NeuroSync AI

1. Project Overview

NeuroNest AI is an AI-powered web-based platform designed to improve mental well-being, emotional health, productivity, and healthy lifestyle management for users of all age groups. It uses Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and emotion recognition technologies to analyze user data such as mood, sleep patterns, daily activities, and behavioral trends. Based on this analysis, the system provides personalized wellness recommendations, emotional support, productivity guidance, and healthy lifestyle suggestions. It also includes features like task management, reminders, goal tracking, gamification, and parent-managed baby profiles, creating a secure and intelligent wellness ecosystem for individuals and families.

2. System Scope

The proposed system is developed as an **academic project** that can later be expanded into a full-scale implementation.

Current scope includes:

- Student mental health assessment
- Mood tracking
- Stress evaluation
- AI-based recommendation engine
- Parent monitoring
- Administrative management
- Facial emotion recognition
- Voice emotion analysis

Future enhancements may include:

- Smart wearable integration
- Mobile application
- Predictive mental health analytics
- Emergency alert system
- Counselor support

3. Target Audience

The primary users of the system include:

- Students
- Parents
- System Administrators

4. User Roles

- Administrator – Manages users, monitors the system, generates reports, and configures AI settings.
- Registered User – Tracks emotional health, lifestyle, productivity, goals, and receives AI-based recommendations.
- Student – Monitors study habits, tracks stress and focus levels, and receives AI-based study and productivity recommendations.
- Parent/Guardian – Manages baby or child profiles, monitors health and development, and receives AI parenting suggestions.

5. Modules

1. User Management Module – Registration, login, profile, and role management.
2. Parent & Baby Care Module – Baby profiles, growth, sleep, feeding, and health tracking.
3. Emotion Recognition Module – Mood tracking and emotion analysis.
4. Mental Wellness Module – Stress management and wellness support.
5. Lifestyle Management Module – Sleep, activity, and healthy habit tracking.
6. Productivity & Goal Management Module – Tasks, goals, reminders, and progress tracking.
7. AI Recommendation Module – Personalized wellness and lifestyle suggestions.
8. AI Chat Assistant Module – AI-powered emotional support and guidance.
9. Gamification Module – Badges, rewards, challenges, and activity streaks.
10. Reports & Analytics Module – Wellness reports and progress insights.
11. Notification Module – Reminders and motivational alerts.
12. Admin Dashboard Module – System monitoring, user management, and analytics.

6. System Ownership

The proposed system is developed as an **academic project** under the Department of Computer Applications.

System Owner:

Academic Institution (Amal Jyothi college of engineering kanjirapally)

Developed by, Krishnapriya Rajesh (Final Year MCA Students) Future ownership may be transferred to educational institutions for practical implementation.

7. Questionnaire For Data Collection

Target Users: Parents

- 1. How often do you monitor your child's daily activities?**
Daily.
- 2. What is your biggest concern about your child's well-being?**
Managing emotional health, sleep, and screen time.
- 3. Do you maintain records of your child's sleep, meals, and health?**
Sometimes, but it is difficult to manage consistently.
- 4. Would you like to receive AI-based recommendations for your child's health and development?**
Yes, personalized suggestions would be helpful.
- 5. Which feature would you find most useful?**
Sleep tracking, growth monitoring, and health reminders.
- 6. How do you currently manage vaccination or medical reminders?**
Using phone reminders or a personal diary.
- 7. Do you think a single platform for monitoring your child's health and activities would be beneficial?**
Yes, it would make monitoring easier.
- 8. Would you recommend such a platform to other parents?**
Yes.

Target Users: Students

- 1. Do you often experience stress due to studies or examinations?**
Yes.
- 2. How many hours do you study on an average day?**
Around 2–4 hours.
- 3. Do you find it difficult to concentrate while studying?**
Sometimes.
- 4. Would AI-based study recommendations and productivity tips help you?**
Yes.

5. What is the biggest reason for losing focus during studies?

Mobile phone distractions and social media.

6. Do you regularly prepare a study schedule?

Occasionally.

7. Which feature would help you the most?:

Study planner, reminders, and focus tracker.

8. Would you like an AI assistant that provides emotional support and motivation?

Yes.

9. Do rewards and achievement badges motivate you to complete tasks?

Yes.

10. Would you use a platform that helps improve both mental well-being and academic productivity?

Yes.

8. Data Collection Contact

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